

Re-normalization and the functional form

Lessons from Agostinelli and Wiswall (2025)

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Where We Are Going

A short review:

1. Lecture 1: a dynamic skill-formation technology
2. Main idea 1: we care about growth or decline of skills
3. Measurement restriction should be age-invariant
4. Main idea 2: CES and its implications on the dynamic skill formation
5. An application using CNLSY79

Three Policy Questions

1. When is investment most productive?

$$\frac{\partial \theta_T}{\partial I_t}$$

어느 시점의 투자가 최종 능력에 가장 큰 영향을 미치는가?

2. Who benefits most from investment?

$$\frac{\partial^2 \theta_{t+1}}{\partial \theta_t \partial I_t}$$

능력 수준에 따라 투자의 효과가 달라지는가?

3. How do early and later investments interact?

$$\frac{\partial^2 \theta_T}{\partial I_t \partial I_{t+1}}$$

초기 투자가 후속 투자의 생산성을 높이는가?

The Dynamic Skill-Formation Model

For a child i , let θ_{it} denote the stock of skills and l_{it} the flow of parental investment.

$$\theta_{i,t+1} = f(\theta_{it}, l_{it}, \eta_{it}), \quad t = 1, 2, \dots, T.$$

- ▶ θ_{it} is a state variable: it carries information from the past.
- ▶ l_{it} is a current input chosen by the family.
- ▶ η_{it} collects shocks and omitted inputs.
- ▶ The technology can change with developmental stage t .

AW's Main Point

We should not fix the scale for different time periods

- ▶ Test scores have arbitrary locations and scales
- ▶ A test may become more sensitive to a latent skill as children age
- ▶ If the measurement scale changes with age, apparent skill growth can be confused with a change in the technology.
- ▶ Therefore, age-by-age normalization can change the object being compared.

(Review) We Observe Measures, Not Skills

A test score is a noisy measure of a latent skill.

$$Z_{t,m} = \mu_{t,m} + \lambda_{t,m} \ln \theta_t + \epsilon_{t,m}, \quad m = 1, \dots, M_t.$$

- ▶ $Z_{t,m}$: observed test or assessment
- ▶ $\mu_{t,m}$: location of the measure
- ▶ $\lambda_{t,m}$: scale / factor loading
- ▶ $\epsilon_{t,m}$: measure-specific noise

Initial-Period Measurement: Three Measures

Suppose three measures are available at the initial period:

$$Z_{0,m} = \mu_{0,m} + \lambda_{0,m} \ln \theta_0 + \epsilon_{0,m}, \quad m = 1, 2, 3.$$

Assume mutually independent measurement errors and normalize $\lambda_{0,1} = 1$.

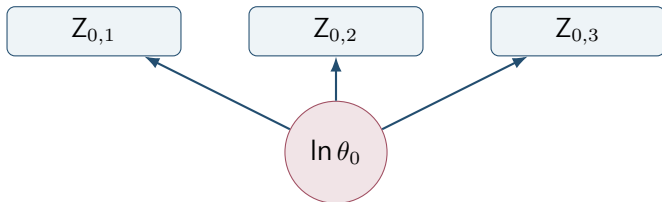
$$C_{12} = \lambda_{0,2} V_0, \quad C_{13} = \lambda_{0,3} V_0, \quad C_{23} = \lambda_{0,2} \lambda_{0,3} V_0,$$

where $V_0 = \text{Var}(\ln \theta_0)$.

$$V_0 = \frac{C_{12} C_{13}}{C_{23}}$$

$$\lambda_{0,2} = \frac{C_{12}}{V_0}, \quad \lambda_{0,3} = \frac{C_{13}}{V_0}$$

What the Covariances Are Doing



common covariance identifies the latent signal

- ▶ C_{12} , C_{13} and C_{23} share the same latent variance.
- ▶ Their ratios recover relative factor loadings.
- ▶ The idiosyncratic parts $\epsilon_{0,m}$ do not generate cross-measure covariance.

The Problem Begins After Period 1

Without further restrictions, the later-period measurement equation is

$$Z_{t,m} = \mu_{t,m} + \lambda_{t,m} \ln \theta_t + \epsilon_{t,m}.$$

Even if later-period covariances identify the factor up to a scale, that scale need not be the same as at period 1.

$$\widetilde{\ln \theta_t} = a_t + b_t \ln \theta_t, \quad b_t > 0.$$

- ▶ Each t can have its own location a_t and scale b_t .
- ▶ The production function and the measurement equation can absorb this transformation.
- ▶ Therefore, latent skill growth and technology parameters are not separately determined.

시점별로 factor loading을 다시 1로 만드는 것은 단순한 단위 선택이 아니라, 시점 간 비교에 제약을 추가할 수 있음

Why Re-normalization Changes the Question

Consider the observed mean change:

$$\begin{aligned} E(Z_{t+1,m}) - E(Z_{t,m}) &= (\mu_{t+1,m} - \mu_{t,m}) \\ &\quad + (\lambda_{t+1,m} - \lambda_{t,m})E(\ln \theta_t) \\ &\quad + \lambda_{t+1,m} [E(\ln \theta_{t+1}) - E(\ln \theta_t)] . \end{aligned}$$

The observed change can reflect:

1. a change in the mean of the measure;
2. a change in the sensitivity of the measure to skill;
3. genuine growth in latent skill.

$\lambda_{t+1,m}$ 이 Key Parameter. 매 t마다 normalization하면
 $\lambda_{t+1,m}$ 의 식별이 혼란스러워짐

The Same Problem for Investment Productivity

The observed sensitivity of a measure to investment is

$$\frac{\partial E(Z_{t+1,m} \mid I_t)}{\partial I_t} = \lambda_{t+1,m} \frac{\partial E(\ln \theta_{t+1} \mid I_t)}{\partial I_t}.$$

Comparing two ages gives

$$\begin{aligned} & \frac{\partial E(Z_{t+1,m} \mid I_t)}{\partial I_t} - \frac{\partial E(Z_{t,m} \mid I_{t-1})}{\partial I_{t-1}} \\ &= \lambda_{t+1,m} \frac{\partial E(\ln \theta_{t+1} \mid I_t)}{\partial I_t} - \lambda_{t,m} \frac{\partial E(\ln \theta_t \mid I_{t-1})}{\partial I_{t-1}}. \end{aligned}$$

- ▶ A changing observed sensitivity need not imply a changing primitive return.
- ▶ This is why “critical periods” require cross-age measurement restrictions.

Age-Invariant Measures

Definition

A pair of measures $Z_{t,m}$ and $Z_{t+1,m}$ is age-invariant if

$$E(Z_{t,m} \mid \theta_t = p) = E(Z_{t+1,m} \mid \theta_{t+1} = p) \quad \text{for all } p.$$

Under the linear measurement model, this implies

$$\mu_{t,m} = \mu_{t+1,m} = \mu_m, \quad \lambda_{t,m} = \lambda_{t+1,m} = \lambda_m.$$

- ▶ The same latent skill level has the same expected score at different ages.
- ▶ The measurement scale is linked across periods.
- ▶ The production technology can vary with age without being confounded with the scale.

Age-Invariant Does Not Mean Age-Standardized

- ▶ It does not require that all children answer the same questions.
- ▶ It does not mean subtracting the age-specific mean and dividing by the age-specific standard deviation.
- ▶ It does not require the skill distributions to overlap across ages.
- ▶ It means that the measurement relationship is stable for a common latent skill level.

Example

PIAT Mathematics is designed to measure achievement over a broad age range. AW argue that its scale can plausibly be treated as age-invariant from ages 5–14.

중요한 것은 모든 측정치가 age-invariant일 필요는 없다는 점이다. 적어도 하나의 적절한 연결고리가 있으면 된다.

CHS and AW: What Is Different?

	CHS	Agostinelli–Wiswall
Normalization	Factor scale is fixed period by period.	Initial-period normalization; later periods are linked by measurement restrictions.
Cross-age scale	Not separately emphasized in the standard implementation.	Requires an age-invariant measure or an alternative production restriction.
Technology	CES imposes weakly positive static complementarity.	Translog interaction allows positive or negative static complementarity.
Policy interpretation	Changes in estimated investment coefficients are read through the imposed scale.	Age comparisons are interpreted only after identifying the common measurement scale.
Main lesson	Flexible measurement, restrictive technology.	More restrictive measurement, flexible technology.

Restrictions can be moved between the measurement model and the technology, but they cannot be ignored.

AW's Main Idea 2: Trans-log

Agostinelli–Wiswall use a flexible translog specification:

$$\ln \theta_{t+1} = \ln A_t + \gamma_{1,t} \ln \theta_t + \gamma_{2,t} \ln l_t + \gamma_{3,t} \ln l_t \ln \theta_t + \eta_{\theta,t}$$

The investment elasticity is

$$e_t^l(\theta_t) \equiv \frac{\partial \ln \theta_{t+1}}{\partial \ln l_t} = \gamma_{2,t} + \gamma_{3,t} \ln \theta_t.$$

- ▶ $\gamma_{1,t}$: self-productivity in logs
- ▶ $\gamma_{2,t}$: investment elasticity at $\ln \theta_t = 0$
- ▶ $\gamma_{3,t}$: heterogeneity in investment productivity

The CHS Benchmark

The CES technology used in much of the earlier literature can be written as

$$\theta_{t+1} = [\gamma_{1,t}\theta_t^{\varphi_t} + \gamma_{2,t}l_t^{\varphi_t}]^{1/\varphi_t}.$$

With positive input weights and $\varphi_t \leq 1$,

$$\frac{\partial^2 \theta_{t+1}}{\partial \theta_t \partial l_t} \geq 0.$$

- ▶ Current skill and current investment are weak complements.
- ▶ Investment is mechanically more productive for higher-skilled children.
- ▶ However, this does not rule out early intervention being optimal through dynamic effects.

What Does the Interaction Coefficient Mean?

$$\frac{\partial e_t^l(\theta_t)}{\partial \ln \theta_t} = \frac{\partial^2 \ln \theta_{t+1}}{\partial \ln \theta_t \partial \ln l_t} = \gamma_{3,t}.$$

Interpretation

- ▶ $\gamma_{3,t} > 0$: direct/log complementarity; investment is more productive for high-skill children.
- ▶ $\gamma_{3,t} < 0$: compensatory investment; investment is more productive for low-skill children.
- ▶ $\gamma_{3,t} = 0$: the investment elasticity is homogeneous in skill.

여기서 부호는 수준(level)의 단순한 cross-partial이 아니라, 로그 능력의 투자 탄력성이 현재 능력에 따라 어떻게 변하는지를 나타낸다.

Static and Dynamic Complementarity

Static / direct

Does current skill change the productivity of current investment?

$$\frac{\partial^2 \ln \theta_{t+1}}{\partial \ln \theta_t \partial \ln l_t} = \gamma_{3,t}$$

Dynamic

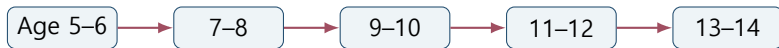
Does early investment change the productivity of later investment through the skill stock it creates?

$$\frac{\partial}{\partial l_t} \left(\frac{\partial \theta_T}{\partial l_{t+1}} \right)$$

- ▶ A negative static complementarity does not imply a negative dynamic effect.
- ▶ Early investment can still raise later skills and later investment returns.
- ▶ The two concepts answer different policy questions.

AW Application: Data and Developmental Stages

- ▶ **Data:** Children and Young Adults of the NLSY79 (CNLSY79)
- ▶ **Sample:** 11,509 children in the main application
- ▶ **Initial period:** ages 5–6
- ▶ **Final childhood period:** ages 13–14
- ▶ Four two-year transitions: 5–6 → 7–8 → 9–10 → 11–12 → 13–14



The data follow children from age 5–6 through adulthood using the NLSY79 Child and Young Adult surveys.

What Is Measured in the Application?

Latent object	Observed measures	Role
Child cognitive skill	PIAT Mathematics, Reading Recognition, Reading Comprehension, PPVT	Multiple noisy measures; PIAT Mathematics is the age-invariant anchor.
Parental investment	HOME score items, reading, praise, museum visits, and related measures	Latent quantity and quality of parental investment.
Mother's cognitive skill	ASVAB subscales	Initial condition and predictor of investment.
Mother's noncognitive skill	Rotter and Rosenberg indexes	Initial condition and predictor of investment.
Adult outcomes	Highest grade completed at age 23 or older; earnings at age 29	Economically interpretable validation and policy metrics.

측정모형은 아동의 능력뿐 아니라 투자와 어머니의 능력도 latent하게 처리한다.

The Age-Invariant Anchor in Practice

For PIAT Mathematics, AW impose a common measurement relationship:

$$Z_{t,\text{Math}} = \mu_{\text{Math}} + \lambda_{\text{Math}} \ln \theta_t + \epsilon_{t,\text{Math}}, \quad t = 0, 1, 2, 3, 4.$$

The remaining measures help recover the latent factor and measurement error:

$$Z_{t,m} = \mu_{t,m} + \lambda_{t,m} \ln \theta_t + \epsilon_{t,m}.$$

- ▶ PIAT Math links the skill unit across ages.
- ▶ Recognition and Comprehension provide additional information and instruments.
- ▶ The measurement relationship is an assumption about the test, not a claim that children stop developing.

A Signal-and-Noise Reading of the Measures

For a measure $Z_{t,m}$, the signal share is

$$\text{SignalShare}_{t,m} = \frac{\lambda_{t,m}^2 \text{Var}(\ln \theta_t)}{\lambda_{t,m}^2 \text{Var}(\ln \theta_t) + \text{Var}(\epsilon_{t,m})}.$$

Age	Measure	Signal share
5–6	PIAT Mathematics	0.273
7–8	PIAT Mathematics	0.764
9–10	PIAT Mathematics	0.777
11–12	PIAT Mathematics	0.804
13–14	PIAT Mathematics	0.924

- ▶ At the initial age, the normalized Math measure contains substantial noise.
- ▶ The same measure becomes more informative at later ages.
- ▶ This is exactly why the loading must be linked across ages rather than reset mechanically.

Signal shares are reported from the measurement estimates in the AW appendix.

Normalization Changes the Skill Unit

Let $\ell_t \equiv \ln \theta_t$.

Cross-age normalization: AW

$$E(\ell_0) = 0, \quad \lambda_{0,m^*} = 1,$$

and for an age-invariant measure m^* ,

$$\mu_{t,m^*} = \mu_{0,m^*}, \quad \lambda_{t,m^*} = \lambda_{0,m^*} \quad \text{for every } t.$$

$$\tilde{\ell}_t = a_t + b_t \ell_t$$

If b_t varies with age, changes in observed scores can be confused with changes in the skill-formation technology.

- ▶ The steps use alternative measures as instruments for measurement error.
- ▶ The main production function is estimated after constructing latent-variable residual measures.
- ▶ Counterfactuals require the joint distribution of initial skills, investment, and income.

Investment Function: Kept in the Background

AW model parental investment as

$$\ln I_t = \alpha_{1,t} \ln \theta_t + \alpha_{2,t} \ln \theta_{MC} + \alpha_{3,t} \ln \theta_{MN} + \alpha_{4,t} \ln Y_t + \eta_{I,t}.$$

- ▶ Investment responds to the child's skills, maternal skills, and income.
- ▶ $\alpha_{1,t} > 0$ is reinforcement; $\alpha_{1,t} < 0$ is compensation in the investment choice equation.
- ▶ This is a reduced-form approximation to parental behavior, not a complete household-choice model.
- ▶ The investment equation matters because I_t is endogenous and measured with error.

Why OLS Is Not Enough

Suppose we use a noisy measure of skill in the production equation:

$$\widetilde{\ln \theta_t} = \ln \theta_t + e_{\theta,t}, \quad \widetilde{\ln l_t} = \ln l_t + e_{l,t}.$$

Then the observed regressors are correlated with the regression residual:

$$\text{Cov}(\widetilde{\ln \theta_t}, u_t) \neq 0, \quad \text{Cov}(\widetilde{\ln l_t}, u_t) \neq 0.$$

- ▶ OLS conflates latent skill with measurement noise.
- ▶ The interaction term is especially sensitive to this problem.
- ▶ Alternative measures provide excluded instruments under AW's independence assumptions.

Step 3: Estimate the Skill Technology with IV

Construct a residual measure of the latent variable:

$$\tilde{Z}_{\theta,t,m} \equiv \frac{Z_{\theta,t,m} - \mu_{t,m}}{\lambda_{t,m}} = \ln \theta_t + e_{\theta,t,m}.$$

Substituting noisy measures into the translog technology gives a reduced form:

$$\begin{aligned} Z_{\theta,t+1,m} = & \delta_{0,t,m} + \delta_{1,t,m} \tilde{Z}_{\theta,t,m'} + \delta_{2,t,m} \tilde{Z}_{l,t,m'} \\ & + \delta_{3,t,m} \tilde{Z}_{\theta,t,m'} \tilde{Z}_{l,t,m'} + \pi_{t,m}. \end{aligned}$$

$$\gamma_{j,t} = \frac{\delta_{j,t,m}}{\lambda_{\theta,t+1,m}}, \quad j = 1, 2, 3$$

다른 test와 investment measure를 instrument로 사용해 measurement error를 분리한다.

What Identifies the Interaction?

The interaction coefficient is identified from variation in

$$\tilde{Z}_{\theta,t,m'} \times \tilde{Z}_{l,t,m'}.$$

The excluded instruments include

$$\tilde{Z}_{\theta,t,m''}, \quad \tilde{Z}_{l,t,m''}, \quad \tilde{Z}_{\theta,t,m''} \times \tilde{Z}_{l,t,m''}.$$

- ▶ The alternative measures share the same latent variables.
- ▶ Their measurement errors are assumed independent across measures.
- ▶ The factor loadings must already be known or identified.
- ▶ The age-invariant restriction supplies the cross-age scale needed for $\lambda_{t+1,m}$.

Measurement restrictions make a flexible technology estimable.

AW Estimates: Skill Technology

	5–6	7–8	9–10	11–12
Log skills $\gamma_{1,t}$	1.966 (0.153)	1.086 (0.035)	0.897 (0.027)	1.065 (0.029)
Log investment $\gamma_{2,t}$	0.799 (0.261)	0.695 (0.337)	0.713 (0.403)	0.252 (0.538)
Log skills $\times$ log investment $\gamma_{3,t}$	−0.105 (0.066)	−0.005 (0.019)	−0.003 (0.013)	0.003 (0.010)
Return to scale (at sample means)	2.660	1.776	1.606	1.320
Standard deviation of shocks	5.612	4.519	3.585	4.019

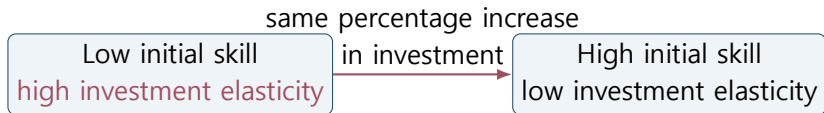
- ▶ Skills are self-productive and investment raises next-period skill.
- ▶ The early interaction estimate is negative: the investment elasticity falls with current skill.
- ▶ The interaction is close to zero in later periods.

Estimates from AW Table 2; standard errors are in parentheses.

Reading the Early Interaction Coefficient

At ages 5–6,

$$e_0^l(\theta_0) = 0.799 - 0.105 \ln \theta_0.$$



- ▶ For low-skill children, the investment elasticity approaches about 1.5.
- ▶ For high-skill children, it approaches about 0.2.
- ▶ This is compensatory static complementarity, not a claim that investment is useless for high-skill children.

Early Investment and Skill Persistence

The estimated technology implies that early inputs matter through two channels:

$$l_0 \longrightarrow \theta_1 \longrightarrow \theta_2 \longrightarrow \theta_3 \longrightarrow \theta_4.$$

- ▶ The age 5–6 production function has high self-productivity and high investment productivity.
- ▶ The estimated total factor productivity falls sharply by ages 11–12.
- ▶ Early investment can therefore have a large effect on final skills even if later investment remains important.

Careful interpretation

The age-invariant measure is what allows AW to compare these coefficients across stages. Without it, a larger early coefficient could partly reflect a different measurement scale.

Adult Outcomes as an Economic Metric

AW link final latent skills to adult outcomes:

$$S_{23} = \mu_S + \alpha_S \ln \theta_4 + \eta_S,$$

$$\ln W_{29} = \mu_W + \alpha_W \ln \theta_4 + \eta_W.$$

Their estimates imply approximately

$$\frac{\partial S_{23}}{\partial \ln \theta_4} \approx 0.15 \text{ years of schooling}, \quad \frac{\partial \ln W_{29}}{\partial \ln \theta_4} \approx 0.041.$$

- ▶ Skill units are still constructed, but their consequences are economically interpretable.
- ▶ The adult equations help translate skill changes into policy outcomes.
- ▶ Schooling is observed with less censoring than realized earnings.

Policy Experiment: A \$10,000 Income Transfer

		5–6	7–8	9–10	11–12
Measurement-error	cor-	0.300	0.172	0.129	0.050
	rected				
Not corrected		0.062	0.029	0.022	0.013

- ▶ Entries are additional months of completed schooling at age 23.
- ▶ The same transfer has the largest estimated effect when given at age 5–6.
- ▶ Ignoring measurement error reduces the estimated age 5–6 effect by almost a factor of five.

AW Table 3, Panel A. The transfer is simulated through the estimated investment and skill-formation system.

Who Benefits from the Transfer?

Low-income families

- ▶ The age 5–6 transfer raises completed schooling by about one month or more for the lowest income percentiles.
- ▶ The effect is close to zero for the richest families.

Low-skill children

- ▶ The transfer effect is also larger for children with low initial skills.
- ▶ The heterogeneity is smaller than the heterogeneity by family income.

Targeting by both income and initial skill is most informative.

낮은 능력의 아동에게 투자가 더 생산적이라는 결과와, 낮은 소득 가구에 소득이전의 효과가 더 크다는 결과는 서로 다른 이질성이다.

What Does AW Change Relative to CHS?

1. Static complementarity

$$\gamma_{3,0} < 0$$

makes investment more productive for low-skill children at age 5–6.

2. Timing

early investment productivity > later investment productivity

after accounting for the common measurement scale.

3. Measurement error

corrected policy effect $\gg$ uncorrected policy effect.

Flexible technology plus explicit measurement restrictions
changes the policy conclusions.

What Remains an Assumption?

- ▶ The log-linear measurement model imposes a cardinal structure on observed measures.
- ▶ At least one measure, especially PIAT Mathematics, is treated as age-invariant.
- ▶ Measurement errors are independent across measures and independent of latent variables in the required moments.
- ▶ The investment equation is a reduced-form approximation to household behavior.
- ▶ Counterfactuals require additional assumptions about distributions and dynamic shocks.

The methodological lesson

The question is not whether assumptions can be avoided. The question is whether they are visible, interpretable, and connected to the policy object.

Takeaways

AW move the frontier from re-normalization to cross-age identification.

1. Multiple measures recover latent factors only up to location and scale.
2. Re-normalizing every period can confound measurement changes with skill dynamics.
3. An age-invariant measure links the latent skill scale across ages.
4. A translog interaction permits compensatory as well as reinforcing investment.
5. In the CNLSY79 application, early investment is especially productive for disadvantaged children.

The empirical content of a skill-formation model lives in both equations: the technology and the measurement system.

Applications: To be Added

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